

Reading spiral structure through dark matter halos

An automated route from simulated gas disks to measurable spiral-arm morphology

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Spiral arms are visible. Measuring them consistently is harder.

Disk galaxies span grand-design, multi-armed, and flocculent morphologies.

Visual classifications do not provide reproducible width, contrast, and winding measurements across hundreds of simulated galaxies.

A scalable measurement must turn visual structure into reproducible quantities.



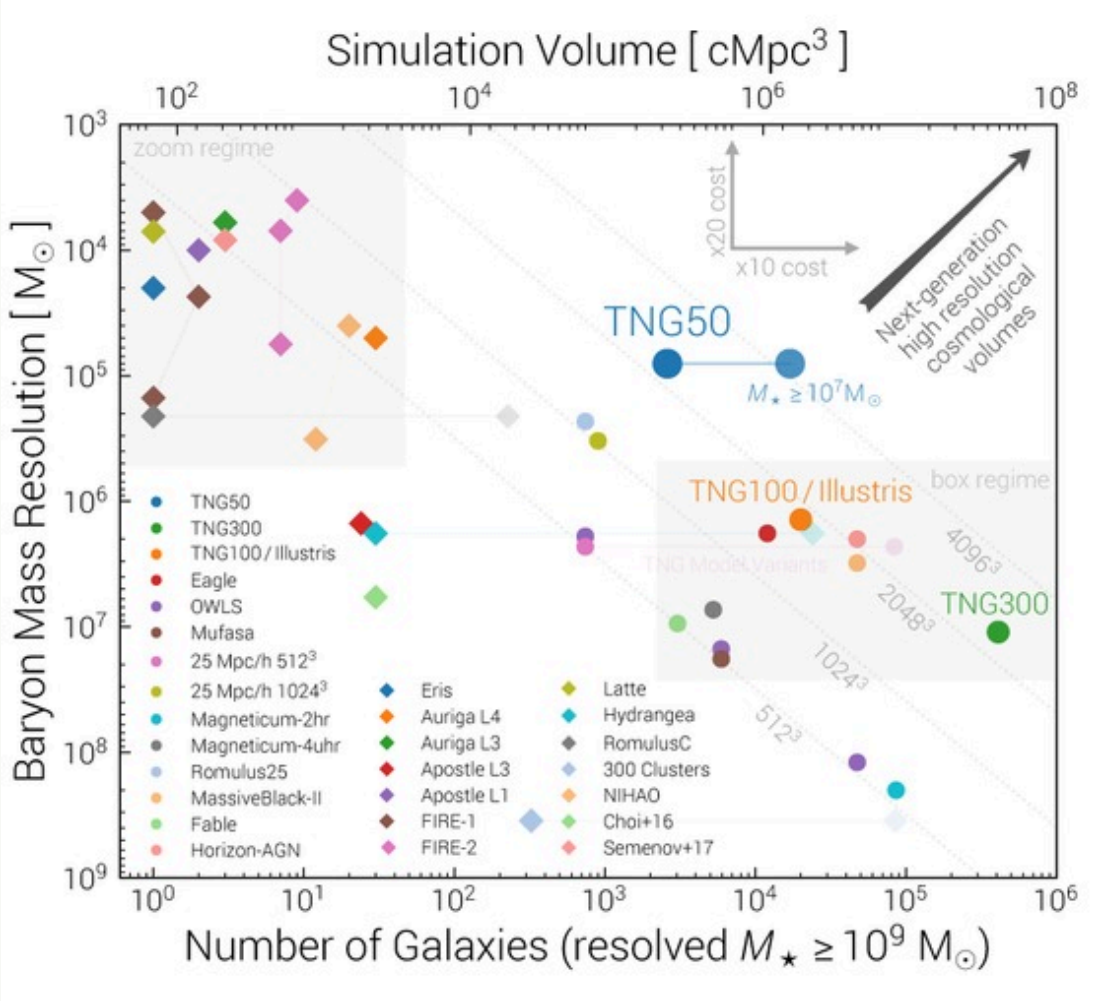
Observed and modeled spiral structure motivates quantitative definitions. Source images compiled in the original presentation.

RESEARCH QUESTION

How do spiral-arm properties relate to the galaxy's host dark matter halo?

Disk-halo correlations can help distinguish among competing scenarios for spiral structure, but only if morphology is measured consistently.

A resolved laboratory: IllustrisTNG50 at $z = 0$



2.7 TB

snapshot
data
analyzed

377

gas-rich
subhalos
preselected

309

disk
systems
passed
filtering

TNG50 combines a cosmological volume with the resolution needed to study galactic structure.

Resolution: $512^3 M_{\odot}$ Simulation: Pillemer et al. (2016)

From a 3D subhalo to a face-on gas disk

1 • Isolate and verify

Group gas spatially with DBSCAN, then visually verify the disk component.

2 • Select disk gas

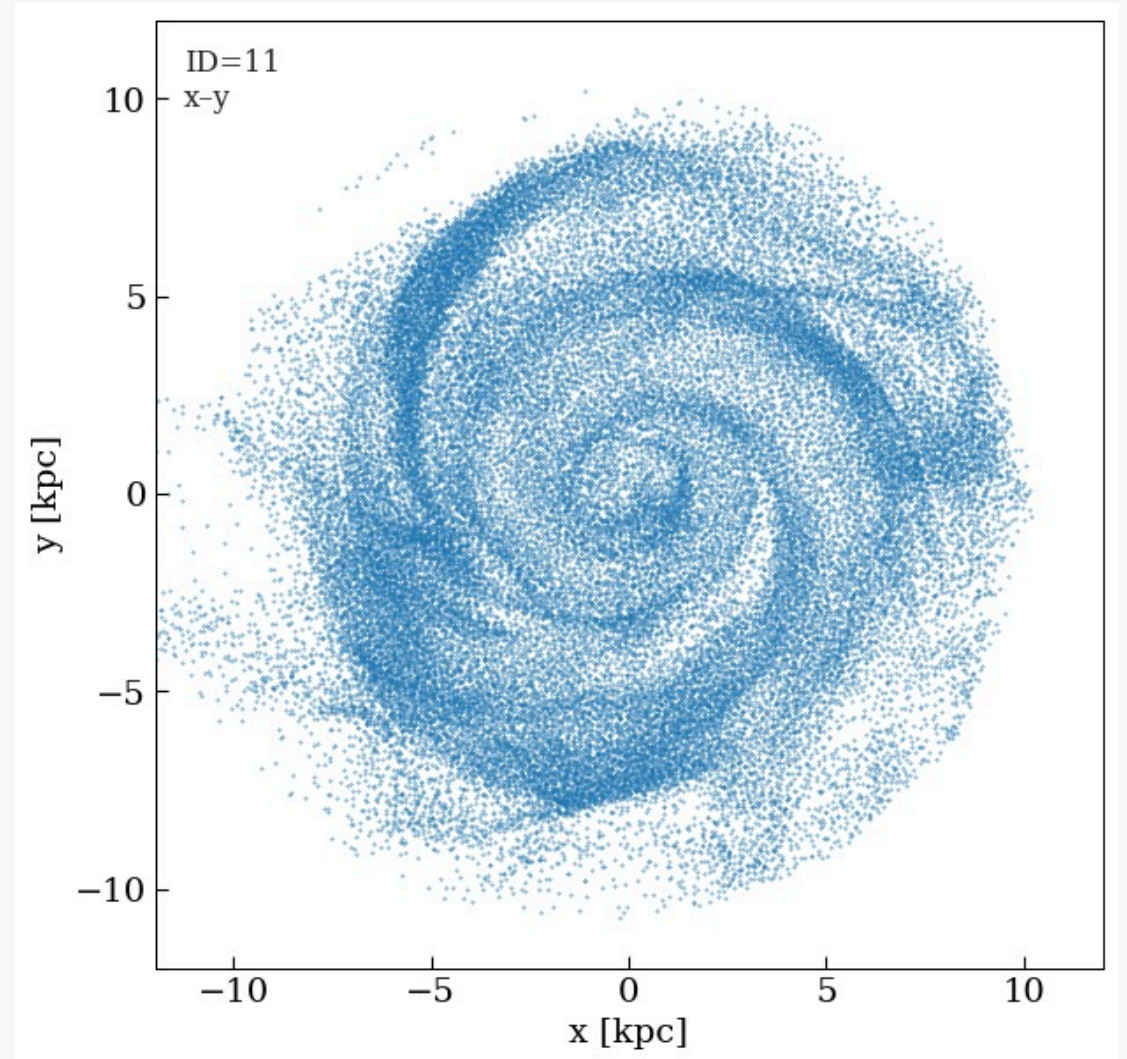
Use internal energy and orbital circularity to retain disk material.

3 • Orient

Align angular momentum

4 • Deproject

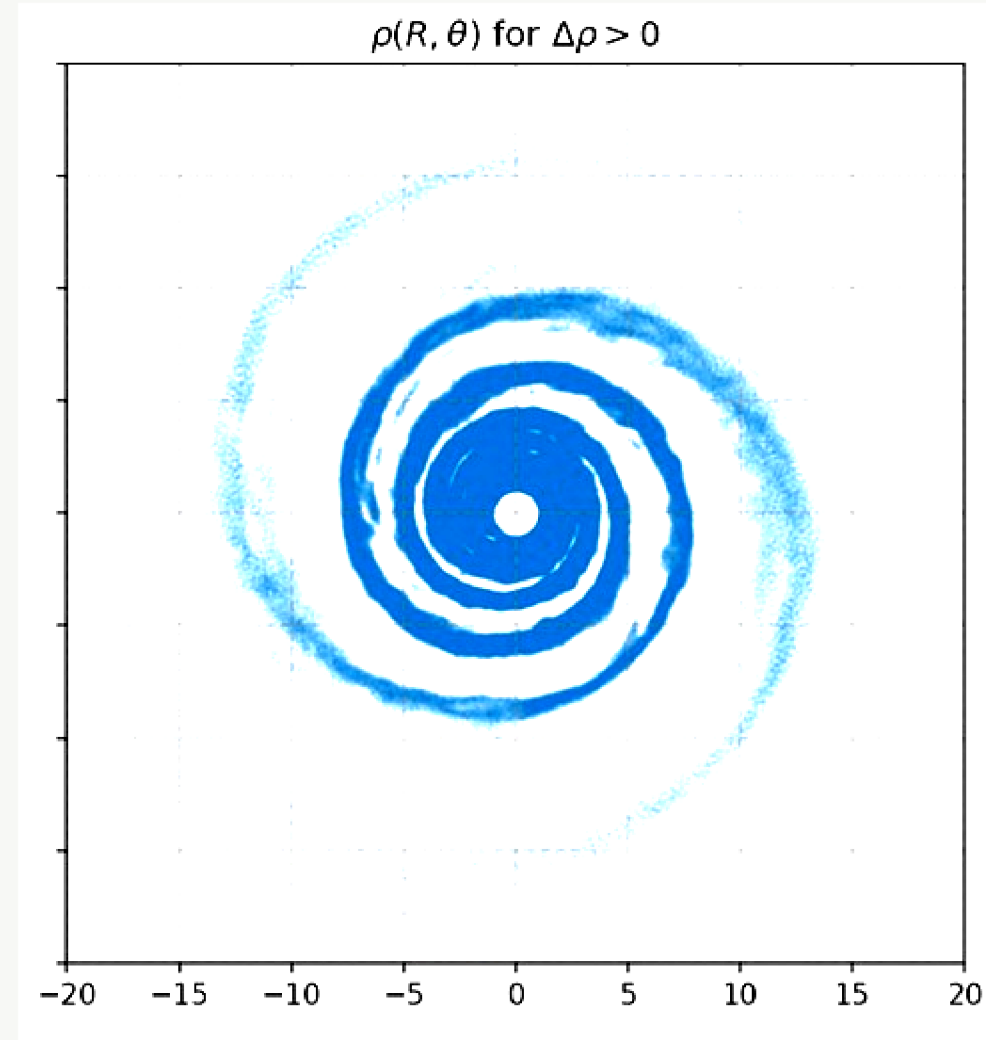
Measure morphology in



Arms emerge as overdensities above the disk profile

1. Estimate the mean radial density profile.
2. Interpolate a smooth background $f(R)$.
3. Compute density contrast relative to that background.
4. Keep connected overdense regions as arm candidates.

This separates “where the disk is dense” from “where density exceeds the local radial expectation.”



Positive density contrast highlights coherent spiral structure.

One pipeline, three morphological measurements

Connected structure

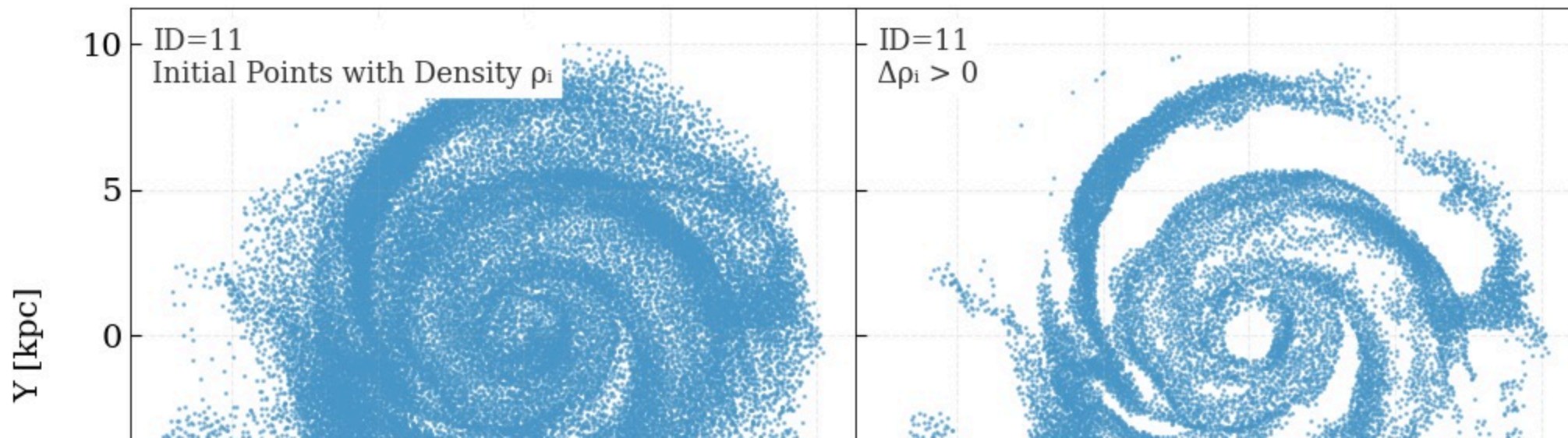
Breadth-first search groups neighboring overdense cells into candidate arms.

Central path

Arm centroids provide a trace through each coherent segment.

Morphology

Transverse profile fits estimate width and amplitude; a polar-plane fit estimates pitch angle.



Method coverage and analysis samples

~80%

of 309 filtered disks
yielded recoverable arms

1,323

arm segments in the full
measurement sample

533

principal arms in a cleaner
267-disk subset

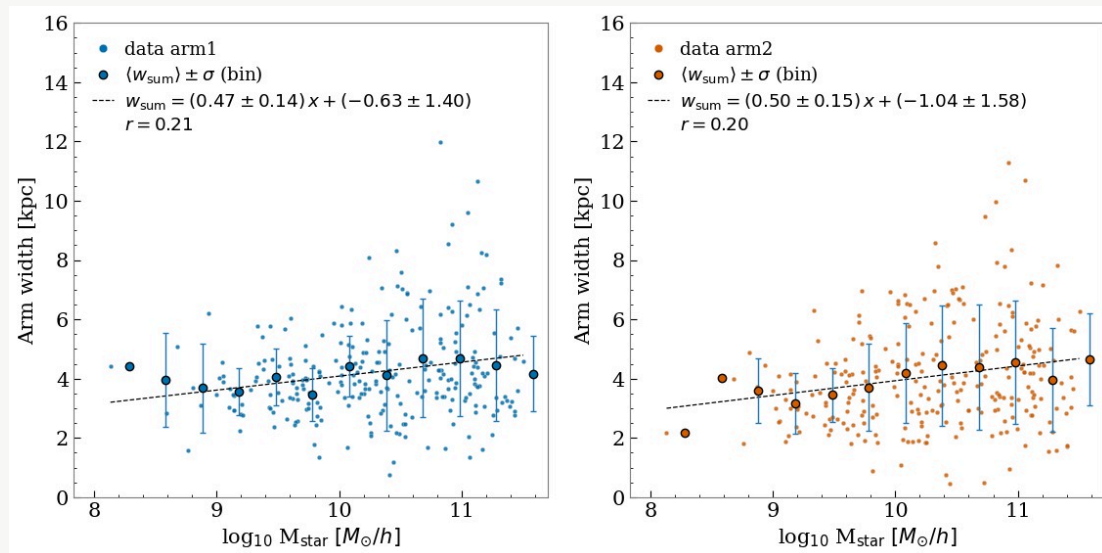
Recovery rate is method coverage, not a scientific correlation.

Failures concentrate in low-density or highly turbulent systems, defining where the current workflow needs improvement.

Arm width tends to increase with galaxy mass

Using the original full-disk estimator, the same qualitative tendency appears against dark matter, gas, and stellar mass.

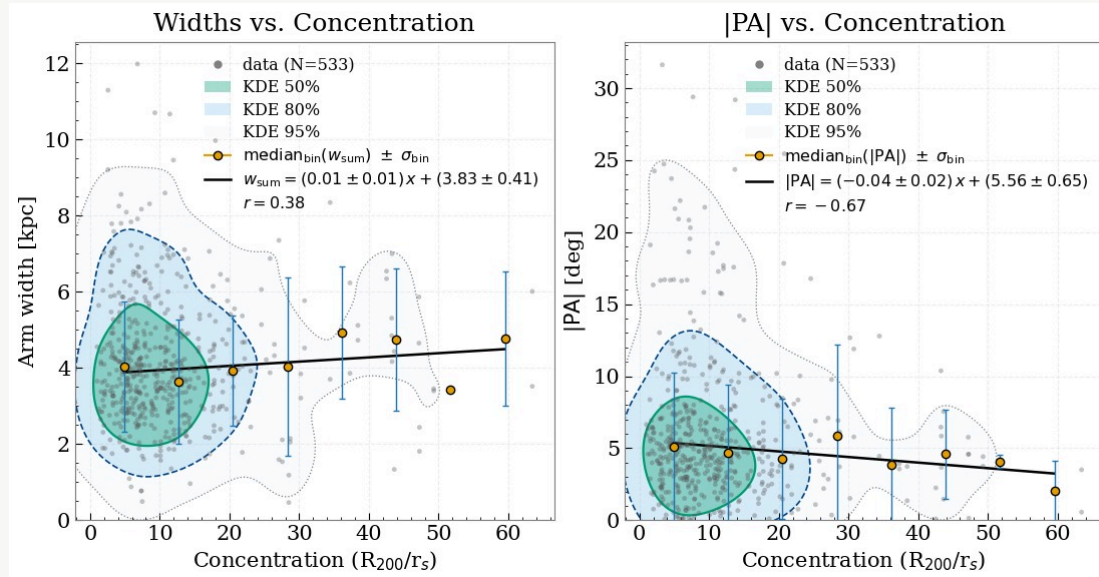
Interpretation: more massive disks may host wider coherent overdensities.



Individual arms are dispersed; binned summaries show a modest upward trend.

Evidence remains weak: fitted slopes differ from zero by no more than about 1.3σ . A group-only estimator removes a known contamination bias, but has not yet been rerun across the full sample.

Pitch angle shows a weak decline with halo concentration



Width is consistent with no concentration trend: 0.011 ± 0.011 kpc per unit of c . Absolute pitch angle declines mildly.

Halo concentration describes how centrally the halo mass is distributed. More concentrated halos are associated with slightly more tightly wound arms in this sample.

Working hypothesis: a more centrally concentrated potential may contribute to the trend.

Association does not establish mechanism. The gravitational explanation remains a hypothesis to test.

What we learned

1. **Most measurement stages are automated.** After disk selection and quality control, the workflow extracts coherent spiral structure reproducibly.
2. **Width carries a tentative mass signal.** Mean arm width rises modestly across multiple mass tracers.
3. **No systematic amplitude–halo–mass trend was detected.** Pearson coefficients are small in this sample ($r \approx 0.06\text{--}0.14$).
4. **A weak pitch–angle–concentration anti-correlation appears.** It requires confirmation and does not establish a mechanism.

What comes next

Method

- Re-run the revised width estimator across the full sample.
- Quantify completeness and uncertainty by morphology.
- Test robustness to selection and clustering choices.

Science

- Expand beyond the current disk sample.
- Compare synthetic and observed arm measurements.
- Separate halo effects from baryonic structure and environment.



The result is a measurement framework and a set of testable trends, not a final causal model.

References and data

Elmegreen, D. M., & Elmegreen, B. G. (1989). *Arm classifications for spiral galaxies*.

Pillepich, A., et al. (2019). *First results from the TNG50 simulation: the evolution of stellar and gaseous discs across cosmic time*. MNRAS, 490, 3196–3233.

Davis, D. R., & Hayes, W. B. (2014). *SpArcFiRe: Scalable Automated Detection of Spiral Galaxy Arm Segments*. ApJ, 790, 87.

Silva-Villa, E., & Cano Gomez, X. (2022). *NGC 5236's stars as tracers of arms and arm widths in spiral galaxies*. MNRAS, 516, 1212–1226.

IllustrisTNG data: <https://www.tng-project.org/> · FAcOm, Universidad de Antioquia